Student Name: _____

Student Email: _____



Quiz C: Convolutions (CS 7150)
Fall 2026

Instructions

- Write your name and your Northeastern email address on every page.
- You have 15 minutes. Write your answers in the spaces directly after each question.
- For worked questions, draw a single BOX around your final answer, so that graders can tell the difference between your answer and your scratch work.
- Closed book: no notes, books, computers, calculators, or other resources, and no communication with anyone other than the proctor.
- You may not share or discuss these questions with any student who has not yet taken the quiz.

I attest that I have read and will follow the closed-book rules above.

Student signature: _____ Date: _____

1. (2 points) True or false, circle one.

   (a) The number of parameters in a `Conv2d` layer depends on the height and width of the images it is applied to. True False

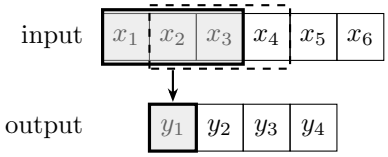
   Solution: False. A convolution has one small kernel per (output channel, input channel) pair, reused at every position, so its parameter count depends only on the channels and the kernel size. The same layer runs on a 32×32 image or a 1000×1000 one.

   (b) Away from the edges, shifting the input of a stride-1 convolution by one pixel shifts its output by one pixel. True False

   Solution: True. This is translation equivariance: every output is computed by the same kernel from the window under it, so moving the input moves the outputs with it. (HW2 found where it breaks: at the edges, where the window runs off the input or into the padding.)

Student Name: _____ Student Email: _____

2. Convolution as a matrix. A `Conv1d` has one input channel, one output channel, a kernel $[w_1, w_2, w_3]$ of size 3, stride 1, and no bias. The input x has length 6. The shaded window is one placement of the kernel, producing $y_1 = w_1x_1 + w_2x_2 + w_3x_3$; then it slides right one step at a time.



Reminder: with no padding and stride 1, a kernel of size K fits in $N - K + 1$ positions of a length-N input; padding with P zeros on each side adds $2P$ more positions.

(a) (1 point) How many outputs are there, with no padding?

Solution: $6 - 3 + 1 = \mathbf{4}$, as drawn.

(b) (3 points) Write the 4×6 matrix A for which $y = Ax$. Its entries are w_1, w_2, w_3, or 0. Hint: convolutions are often called *weight sharing* because many of the entries of A share the same weights.

$A =$

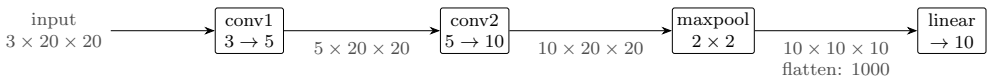
Solution: $A = \begin{bmatrix} w_1 & w_2 & w_3 & 0 & 0 & 0 \\ 0 & w_1 & w_2 & w_3 & 0 & 0 \\ 0 & 0 & w_1 & w_2 & w_3 & 0 \\ 0 & 0 & 0 & w_1 & w_2 & w_3 \end{bmatrix}$, a 4×6 matrix with the kernel shifted one place per row.

Grading: 1 point for the shape 4×6 with one row per output, 1 point for the shifted kernel in every row. Accept the kernel reversed in every row (true convolution rather than the cross-correlation PyTorch computes) if it is consistent.

(c) (1 point) How many zeros of padding on each side keep the output at length 6?

Solution: $6 - 3 + 2P + 1 = 6$ gives $P = \mathbf{1}$.

3. Parameter counting. Images are $3 \times 20 \times 20$. A network is `Conv2d(3, 5, k=3, pad=1)` $\rightarrow$ ReLU $\rightarrow$ `Conv2d(5, 10, k=3, pad=1)` $\rightarrow$ ReLU $\rightarrow$ `MaxPool2d(2,2)` $\rightarrow$ flatten $\rightarrow$ `Linear(1000, 10)`, all with biases.



Reminders: `Conv2d(in, out, k)` has **out** $\times$ **in** $\times$ **k** $\times$ **k** weights and **out** biases. `Linear(in, out)` has **out** $\times$ **in** weights and **out** biases. Pooling and ReLU have no parameters.

(a) (3 points) Parameters in each layer.

conv1 [ ] conv2 [ ] linear [ ]

Solution:

$$\text{conv1} = 5 \times 3 \times 9 + 5 = 135 + 5 = \mathbf{140}$$
$$\text{conv2} = 10 \times 5 \times 9 + 10 = 450 + 10 = \mathbf{460}$$
$$\text{linear} = 10 \times 1000 + 10 = \mathbf{10{,}010}$$

The linear layer is about 94% of the 10,610 total, and the 20×20 image size never enters the convolutions' counts.

Grading: half a point each for the convolutions, 1 point for the linear layer; half credit for each if the weights are right but the biases are left out.